

EPFL

PROCESS BOOK - SPECIAL OPERATIONS

An exploration of European economic progression through interactive data visualization. Built for COM-480 Data Visualization, EPFL.

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1. INTRODUCTION & PROJECT GOAL

For decades, Europe stood as a primary economic powerhouse. Recently, a narrative of European stagnation and relative decline compared to other global regions has emerged. Understanding this trajectory is vital for citizens, policymakers, and entrepreneurs to gauge where global economic weight will sit in the future.

Our project provides an interactive, data-driven analysis to the central question: **"Is Europe really falling behind?"**. To answer this, we contrast macroeconomic indicators (national level) with equity market performance (company level via ETFs and indices) across major global regions from the past few decades.

2. MILESTONE 1

During Milestone 1, we focused on analyzing the top 25 global economies by combining World Bank GDP data with country-specific ETFs. However, our Exploratory Data Analysis (EDA) revealed severe data constraints: ETF data was heavily limited before 1996, and domestic market capitalization data was entirely missing for several major countries in recent years. To overcome these gaps and maintain a complete timeline, we had to include additional data and shift our focus to analyzing the datasets from the year 2000 onward. This strategy allowed us to keep our time-series line charts while expanding our platform to compare big picture data (like GDP and ETFs) directly with the specific breakdown of each country's industrial sectors from the year 2000 onward.

3. MILESTONE 2

For Milestone 2, the initial idea for our graph were the following:

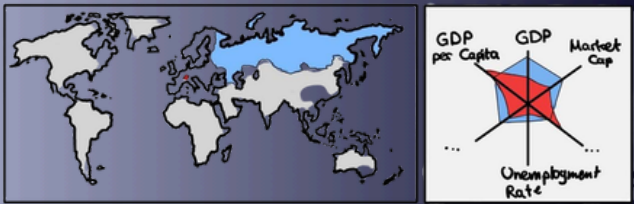


FIG 1. MILESTONE 2 DESIGN DRAFTS FOR SPIDER GRAPH

While the main map shows a broad overview, the **spider graph** enables direct, multi-variable comparisons. Each axis represents a key social or economic indicator (e.g., GDP, Market Cap, Unemployment). By overlaying a few selected countries, users can easily see structural differences and track how economies have evolved over time relative to a chosen reference year.

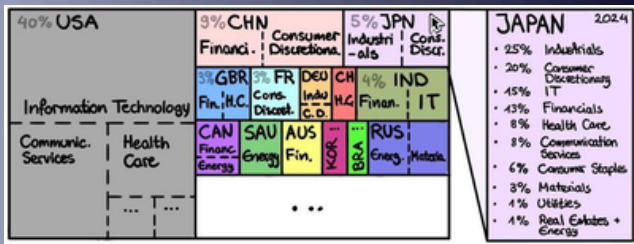


FIG 2. MILESTONE 2 DESIGN DRAFTS FOR SPIDER GRAPH

To strengthen the analysis, we use a tree map to explore a single country's economic evolution broken down by contributing sectors. This comprehensive view helps identify exactly which industries are driving a region's relative economic performance.



FIG 3. MILESTONE 2 DESIGN DRAFTS FOR A CARTOGRAM

We ultimately decided against creating a dynamic cartogram. While it offers an intuitive way to show global shifts by distorting country sizes, it proved redundant for our project since our dataset does not cover all countries.

3. DATA EXPLORATIONS & TECHNICAL CHALLENGES

During our Exploratory Data Analysis (EDA), we uncovered major gaps in domestic total market capitalization data provided by the World Bank. The data was highly fragmented, completely missing recent years for several major global economies.

To maintain a robust visualization, we expanded our dataset. We integrated a third comprehensive dataset containing historical daily data for ETFs corresponding to various countries as well as some global stock indices. While this provided a rich proxy for data on market capitalization for various countries and regions, most such ETFs were established between 1995 and 2000. As such, this limited the complete analysis to the year 2000 and beyond.

Where data points were missing and would disrupt the visualization growth, we interpolated the data points with a local time exponential fit, as all the economic indicators under consideration are expected to undergo exponential growth.

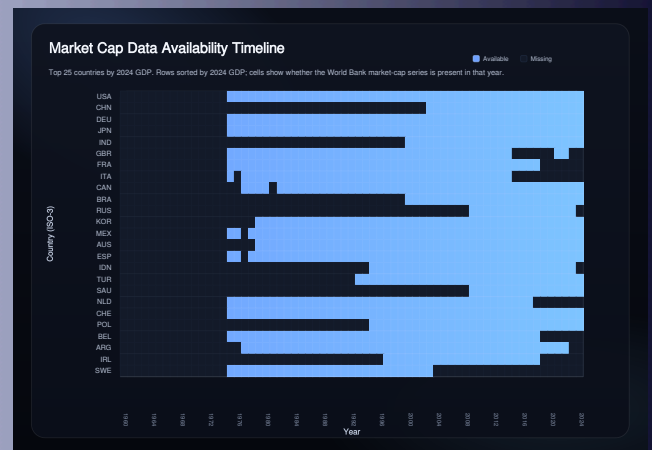


FIG 4. AVAILABILITY OF NATIONAL MARKET CAP DATA FOR TOP 25 COUNTRIES BY GDP. (SOURCE: WORLD BANK)

4. MVP VISUALIZATIONS: CHOROPLETH AND SPIDER PLOT

The central starting figure of our design was a choropleth to visually highlight variations in compounded growth of various factors between regions and countries, and a spider plot to concisely view and compare a selected set of entities across several factors.



FIG 5. CHOROPLETH DISPLAYING A SELECTION OF COUNTRIES' COLOURED BY GDP (RELATIVE TO 2000)

For analyzing properties of the entities across two chosen axes, we also allow switching the choropleth to a scatter plot that evolves with time, and allows seeing the ‘shape’ of a country or region’s development, as compared to its peers or the world at large. This helps in particular to easily visually spot outliers whose economic indicators correlated particularly different from the rest.

The spider plot overlay forms the final piece of the main figure. While the choropleth and scatter plot allow comparison of larger sets of countries or regions across one or two metrics, the spider plot offers the opposite. It allows a deep dive across all available metrics for a small chosen set of countries and observe their development in detail.



FIG 6. SPIDER PLOT OVERLAY SHOWING VISUALIZING A SELECTION OF COUNTRIES’ ECONOMIC METRICS

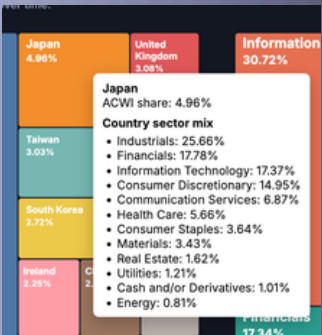


FIG 7. COUNTRY SPECIFIC BREAKDOWN ON HOVER FOR INDUSTRY TREEMAP



FIG 8. SCATTER PLOT SHOWING ETF PRICE PROGRESSION FOR A SELECTION OF EUROPEAN COUNTRIES AND THE US.

5. EVOLUTION OF VISUALIZATIONS: DEEP-DIVE TREEMAP

To address our core question “*Is Europe falling behind?*” we shifted from broad geographical overviews to a coordinated visualization consisting of two side-by-side treemaps. The first treemap displays the global equity weight by country; hovering over any specific country tile reveals its internal sector breakdown. Directly adjacent, the second treemap illustrates the distribution of these sectors on a global scale. Connected by a chronological timeline slider, both treemaps function dynamically to let users seamlessly track how these global structures shifted between 2015 and 2026.

The primary challenge of this phase lay entirely in this dual-layered data collection and harmonization process. To build the country-level treemap, we had to source "dual-layered" factsheets: first extracting the overarching country weights from the MSCI ACWI reports, and then manually gathering individual MSCI Country factsheets for each featured nation to capture their specific sector breakdowns. This complex data aggregation required standardizing sectors across an evolving classification landscape, moving from 10 sectors (2015 - early 2016) to 11 sectors (late 2016 - 2024), and finally to 12 sectors in 2025 and 2026 to incorporate Cash and/or Derivatives. However, once this data was successfully compiled, implementing the visualization was straightforward; our design drafts from Milestone 2 provided a robust blueprint that translated seamlessly into the final D3 interactive layout.



The **MSCI ACWI** (All Country World Index) is a global stock index that tracks nearly 3,000 large and mid-cap companies across developed and emerging markets. Covering about 85% of the world's investable stocks, it is the ultimate benchmark for global equity performance.

The resulting visualization clearly illustrates a widening divergence in global equity concentration and structural composition. Between April 2015 and May 2026, the US expanded its dominant global weight from 52.95% to 63.70%, heavily fueled by an explosive rally in Information Technology, which more than doubled its global sector footprint from 14.54% to 30.72%. In stark contrast, European nations exhibit a fragmented, low-growth homogeneity trapped in traditional, slow-moving sectors like Financials and Industrials. By utilizing the hover feature to analyze individual country profiles, the treemap underscores that Europe is indeed falling behind, not merely due to market size, but because it lacks the high-growth tech engines that are aggressively consolidating global market share in the US.

Based on these treemaps, we added targeted summary tiles at the top to break down what is actually driving this economic divide. These key metrics isolate the main variables to show whether growth in Europe is balanced and widespread, or if it is held back by specific industries. At the same time, they clearly point out the high-growth sectors that are fueling the aggressive economic expansion in the US.

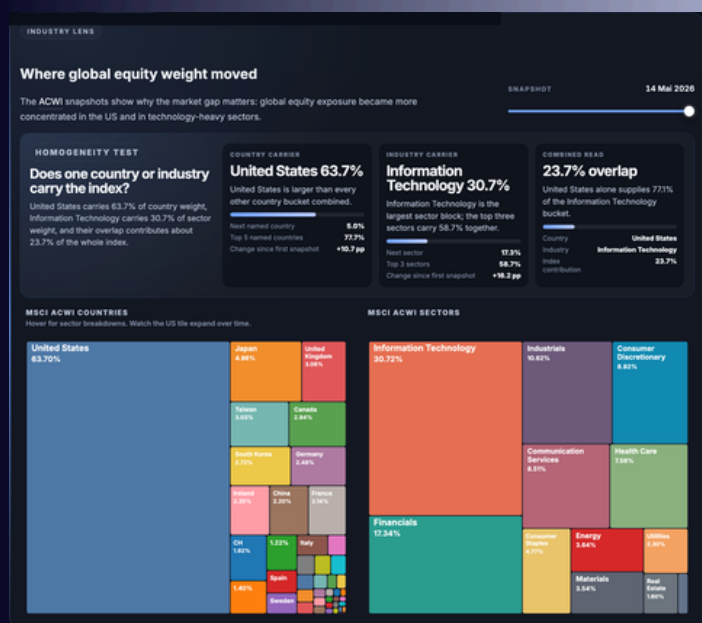


FIG 9. FULL VIEW OF THE TREEMAP SNAPSHOTTED AT EARLY 2026. THE TREEMAP IS ACCOMPANIED BY DYNAMICALLY COMPUTED OVERVIEW OF SELECTED STATISTICS ABOUT THE PROMINENT PLAYERS: UNITED STATES' HEGEMONY AND THE TECH INDUSTRY.

6. EVOLUTION OF VISUALIZATIONS: TIME-SERIES DIVERGENCE PLOT

Our main visualization figures compared relative growths of countries by relativizing their economic metrics to a common year, chosen to be 2000 due to several data points being unavailable before. However, this paints a notably biased picture by relativizing all growth to the year 2000.

This can in particular cause equivalent growth spurts across regions appear very different if their growth patterns had been significantly different before. For example, a 5% YoY growth from 2010 to 2015 would look very different on the plot between two countries that started 2010 at 1x and 3x their year 2000 values for the chosen metric.

To address this and allow more time-local analysis of the data, we made what we call a “time-series divergence plot” or a “normalizing explorer”, dubbed so as its main feature is allowing the flexible choice of comparing different time series together (here for a US/EU pair), and choosing a year to normalize all values against, allowing for much richer time-local analysis of their growth, to understand divergences caused by any common shocks. In particular, this led us to the analysis in the following section.



FIG 10. THE TIME-SERIES DIVERGENCE PLOT, ALLOWING A SELECTION OF CHOICE METRICS. HERE: SHOWING GDP-NOMINAL AND GDP-PPP-ADJUSTED, NORMALIZED AT 2008, DEMONSTRATING A SHARP DIVERGENCE IN NOMINAL FIGURES, WITH A Milder DIVERGENCE IN PPP-ADJUSTED FIGURES SQUEEZED IN.



FIG 11. THE TIME-SERIES DIVERGENCE PLOT, WITH GDP-NOMINAL FIGURES ISOLATED.



FIG 12. DIVERGENCE FIT SCORES FOR YEARS ACROSS THE CHOSEN METRICS. BEST-FITS FOR DIVERGENCE ARE CLUSTERED AROUND 2008.

7. HYPOTHESIS TESTING AND DATA SELECTION CHALLENGES

Our initial analysis and visualizations presented an apparent divergence between the growth of the US when contrasted against the EU as a region. However, this divergence was not easily visible in the initial years of the available data.

This led us to hypothesize the existence of a divergence point, after which the growths stopped tracking each other. We hypothesized this as the years following the 2008 global financial crisis.

We find the best candidate year for divergence by fitting the series (US/EU) pair against a divergent model and ranking them via a Bayesian information criterion test. The analysis confirmed our initial hypothesis with best fits for GDP, ETF prices, and market cap data being 2008-2010.

To make this apparent, we developed the prior time-series divergence plot. However, during data cleanup, we replaced data on nominal GDP by GDP PPP-adjusted, and found that the divergence point had shifted to 2010, and the magnitude of the divergence had shrunk significantly.

This challenged our view of the original hypothesis and led us to consider the viability of economic indicators as proxies of the development and well-being of a populus. We discuss this further in our conclusions.

8. PEER ASSESSMENT & PROJECT BREAKDOWN

To achieve a successful final result, the project responsibilities were distributed equally based on each team member's technical strengths:

Eliota Braha: Worked on the figures of Milestone 2. In charge with the data for the treemaps for the countries and sectors and implemented the treemap section. Improved the webpage and in charge of the process book.

Botond Kovacs: Came up with the general idea of the project, detailed the motivations behind in milestones 1 and 2. Searched for datasets that could be useful. Using AI, coded the MVP of the website, including the main map visualisation, the scatter version, the region grouping. Then iterated on it using team feedback.

Sankalp Gambhir: Performed part of the exploratory data analysis and data preprocessing for milestones 1 and 2. Implemented the spider plot. Designed and implemented the divergence plots and analysis.

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